Normal subgroups

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$N \leq G$ is called normal subgroup of G, denoted as $N \triangleleft G$, if and only if (the following are equivalent):

- $\forall g \in G, gNg^{-1} = N$
- There is some group homomorphism $G \to H$ which has N as kernel.
- It "makes sense" to take quotient of G by N.
- There is some *congruence relation* (i.e., an *equivalence relation* that is compatible with the given algebraic structure (in this case the group structure)) on G such that $N = [i_G]$. (More details in next point).
- N defines a *set* equivalence on G (defined as $g_1 \sim g_2 := g_1 g_2^{-1} \in N$, i.e., N and its cosets form a partition of G). For this equivalence relation to support the algebraic structure (and therefore, become a congruence relation), we should also have that "group products done on equivalent elements should yield equivalent results", i.e, $g_1 \sim g_2, h_1 \sim h_2 \Rightarrow g_1 h_1 \sim g_2 h_2$.

[Sources: Wikipedia, MathStackExchange]

I particularly find the congruence relation idea to be the most intuitive. Let's go through it once more.

I have $N \leq G$, which induces a partition of G into N and its right cosets. This partition corresponds to an *equivalence relation* on G, namely

$$g_1 \sim g_2 := g_1 g_2^{-1} \in N.$$

> Observe that if we had taken left cosets instead, then we'd need to use $g_1^{-1} g_2$.
>
> Why is this so? For left cosets, if g_1 and g_2 belong to the same left coset, then they look like gn_1 and gn_2, so the thing to consider is $g_1^{-1} g_2$. On the other hand, if g_1 and g_2 belong to the same right coset, then they look like $n_1 g$ and $n_2 g$, so we should consider $g_1 g_2^{-1}$.

Now, we want this to also be a *congruence relation* (i.e., the equivalence relations should respect the group structure), therefore we want

$$g_1 \sim g_2 \wedge h_1 \sim h_2 \Rightarrow g_1 h_1 \sim g_2 h_2.$$

> Why have this strange criterion? What we actually want is that the equivalence classes form a group under the induced operation. So, we *want* to define $[g] *_{\text{new}} [h] = [g * h]$, which only makes sense if for any representatives from the 2 equivalence classes, when multiplied together, all the results should lie in the same equivalence class.

Let us show that this is equivalent to the first definition for normality of a subgroup. Given $g \in G, n \in N$, we have $g \sim g$ and $n \sim i_G$, so we get $gn \sim g$, which is the same as $gng^{-1} \in N$. We can use this to prove that gNg^{-1} and N are equal as sets.

Also, suppose $g_1 \sim g_2$ and $h_1 \sim h_2$, so $g_1 g_2^{-1} \in N, h_1 h_2^{-1} \in N$. Using the first definition, we get $g_1 h_1 h_2^{-1} g_1^{-1} \in N$, and right multiplying by $g_1 g_2^{-1}$, we get that $g_1 h_1 h_2^{-1} g_2^{-1} \in N$, i.e., $g_1 h_1 \sim g_2 h_2$.

How does this correspond to how a normal subgroup looks on a Cayley diagram?

See page 132 of Nathan Carter's *Visual Group Theory*, section 7.3 "Normal Subgroups and Quotients".

Another perspective, which is just a re-garbled version of the previous perspective:

Suppose G has elements $\{g_1, g_2, ...\}$, and we have $N \leq G$. Now each element of G corresponds to a left coset, so we have a set map $G \to \{g_1N, g_2N, ...\}$ which maps $g \in G$ to gN.

It is often the case that two or more elements of G correspond to the *same* coset. We want to develop the set of left cosets into a group, so we need to define how to "multiply" two left cosets and get a new one. The "multiplication" we want should behave in the following fashion:

> If g_1 produces a coset L_1 and g_2 produces a coset L_2, then the "product" of L_1 and L_2 should be produced by g_1g_2.

As mentioned in *Artin* (2.12, pg 67), we can think of product of two left cosets L_1 and L_2 as the set $\{l_1l_2 : l_1 \in L_1, l_2 \in L_2\}$. Normality guarantees that product of g_1N and g_2N in this fashion is equal to $(g_1g_2)N$.

However, for that to happen, we need to ensure that:

> If two elements g_1 and ${g_1}'$ produce the same left coset, and g_2 and ${g_2}'$ produce the same left coset, then g_1g_2 and ${g_1}'{g_2}'$ should produce the same left coset.

This is exactly the normality condition on $N \leq G$.

Observe that g_1 and ${g_1}'$ produce the same left coset iff both of them lie in the same left coset. So this is how this perspective links back to the previous one.

Some other/related nice perspectives:

- https://math.stackexchange.com/q/1014535
- https://math.stackexchange.com/q/776039
- https://math.ucr.edu/home/baez/normal.html
- https://gowers.wordpress.com/2011/11/20/normal-subgroups-and-quotient-groups/